Fast and slow neuromodulatory influences

Fast: ACC → locus coeruleus

Level-3 cortical signal → brainstem relay
Phasic NE modulates Level-1 threshold
Tonic NE can alter Level-0 gain
LC is a relay, not a hierarchical stratum

Slow: PVN → HPA axis

CRH → ACTH → cortisol
Rapid membrane-MR facilitation
Sustained GR suppression
Neuroendocrine priors modulate cortex

Coupling notation

$\alpha_{31} = \kappa_{12}\kappa_{23}$ $\alpha_{41} = \kappa_{12}\kappa_{23}\kappa_{34}$
Multi-step influences are compositions, not independent free parameters.

Paper 3 §3 / Fig 3. Subscripted coupling κ differs from the ATP/bit bridge κ.